

# Notes: Bloom (Econometrica, 2009)

## The Impact of Uncertainty Shocks

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## 1 Big picture

- **Research question.** The paper asks what happens to aggregate output, employment, investment, and productivity when uncertainty rises sharply after major political or economic events.
- **Core mechanism.** Firms face nonconvex adjustment costs in labor and capital. When uncertainty rises, the option value of waiting rises. Firms pause hiring and investment until business conditions become clearer.
- **Main result.** An uncertainty shock generates a sharp temporary recession, followed by a rebound and a medium-run overshoot. The recession is short because the shock is second-moment rather than a persistent fall in the level of fundamentals.
- **Empirical discipline.** The paper combines VAR evidence from U.S. macro data with a structural firm-level model estimated using simulated method of moments on Compustat data.
- **Policy message.** The immediate aftermath of an uncertainty shock is a period in which firms are especially insensitive to price changes, so monetary or fiscal policy may have weaker short-run traction.

## 2 Incremental contribution of the paper

- **Second-moment shocks as a recessionary force.** The paper moves beyond first-moment productivity or demand shocks and shows that time-varying uncertainty itself can generate business-cycle dynamics.
- **Structural real-options framework.** The paper embeds stochastic uncertainty into a firm model with labor and capital adjustment frictions, making it possible to simulate the path of the economy after uncertainty jumps.
- **Joint labor-capital adjustment costs.** Instead of estimating investment or labor adjustment costs separately, the paper jointly estimates convex and nonconvex costs for both labor and capital.
- **Link between micro frictions and macro dynamics.** The model uses firm-level evidence on lumpy adjustment to explain aggregate drops and rebounds in employment, output, and productivity.
- **Productivity channel.** The paper emphasizes that uncertainty reduces productivity growth because reallocation from low-productivity to high-productivity units temporarily freezes.

## 3 Paper structure and argument

1. The paper first documents that stock-market volatility spikes after major events and that such spikes are correlated with other uncertainty measures.
2. It then estimates VARs showing that volatility shocks predict a rapid decline and rebound in industrial production and employment.
3. It builds a dynamic firm model in which productivity and demand evolve stochastically and the volatility of that process changes over time.

4. It solves the model numerically and simulates a temporary uncertainty shock.
5. It estimates adjustment-cost parameters using firm-level moments and conducts robustness checks.

## 4 Data and measurement

### 4.1 Uncertainty measurement

- The main uncertainty proxy is stock-market volatility. From 1986 onward, the paper uses the VXO implied volatility index; before 1986, it uses realized monthly volatility of daily S&P 500 returns normalized to the VXO series.
- The volatility shocks are major spikes in the series associated with events such as the Cuban Missile Crisis, JFK assassination, OPEC shocks, Black Monday, the Gulf War, 9/11, World-Com/Enron, and the credit crunch.
- The paper checks that stock-market volatility is correlated with cross-sectional dispersion in firm profits, stock returns, industry TFP growth, and disagreement among macro forecasters.

### 4.2 Macro VAR data

- The baseline VAR uses monthly U.S. data from June 1962 to June 2008.
- Variables include the log S&P 500 index, the volatility shock indicator, Federal Funds Rate, wages, CPI, hours, employment, and industrial production.
- Variables are HP-detrended in the baseline specification. The event indicator for volatility spikes is used to isolate large, plausibly exogenous uncertainty episodes.

### 4.3 Firm-level estimation data

- The structural model is estimated using U.S. firm-level Compustat data.
- The estimation matches moments of investment growth, employment growth, and sales growth, including cross-sectional dispersion, skewness, and dynamic correlations.
- The model explicitly incorporates cross-sectional aggregation across many production units within firms and time aggregation from monthly decision-making to annual accounting data.

## 5 Model

### 5.1 Production and revenue

- A production unit uses capital, labor, hours, and business conditions. Output is Cobb-Douglas, and the firm faces an isoelastic demand curve.
- Productivity and demand are combined into a single “business conditions” state variable.
- Hours are flexible, while labor and capital are costly to adjust.

## 5.2 Stochastic business conditions and uncertainty

- Business conditions contain macro, firm, and unit-level components.
- The standard deviation of business-condition shocks switches between low and high uncertainty states.
- A macro uncertainty shock is modeled as a temporary movement from low to high volatility.

## 5.3 Adjustment costs

- **Partial irreversibility:** hiring/firing costs for labor and resale losses for capital.
- **Fixed disruption costs:** output losses when firms hire, fire, invest, or disinvest.
- **Quadratic costs:** adjustment costs increasing with the rate of labor or capital adjustment.
- These costs create an inaction region. Firms hire or invest only when conditions are sufficiently good, and fire or disinvest only when conditions are sufficiently bad.

# 6 Identification and empirical design

## 6.1 VAR evidence

- The VAR is not the paper’s main structural identification device, but it establishes the empirical shape that the model aims to match: sharp drop, rebound, and overshoot after volatility shocks.
- The comparison with Federal Funds Rate shocks is important. First-moment monetary shocks lead to more persistent declines, while uncertainty shocks generate shorter and sharper cycles.

## 6.2 Structural identification

- The structural estimation identifies adjustment costs by matching moments that distinguish adjustment frictions from persistent firm heterogeneity.
- Investment, employment, and sales growth moments are jointly informative: using only investment or only labor data would make it difficult to separate adjustment costs from the stochastic driving process.
- The estimation is simulated method of moments. Parameters are chosen so that simulated firms match the observed distribution and dynamics of real firms.

# 7 Main findings

## 7.1 VAR results

- Industrial production falls quickly after an uncertainty shock, reaching a trough around four to six months after the shock, and then rebounds.
- Employment shows a similar drop and recovery pattern.
- The estimated path is consistent with the model’s real-options mechanism.

## 7.2 Model simulation results

- Higher uncertainty expands the inaction region. Firms temporarily delay hiring and investment.
- Aggregate labor and output fall immediately after the shock.
- When uncertainty declines, firms address pent-up demand, causing a rebound.
- Because the adjustment function is locally convex near thresholds, the volatility induced by the shock can create a medium-run overshoot.
- Productivity growth falls because reallocation freezes; it rebounds when hiring and investment resume.

### 7.3 Adjustment-cost estimates

- The preferred estimates imply substantial nonconvex capital adjustment costs and meaningful labor adjustment costs.
- Ignoring capital adjustment costs creates substantial bias. Ignoring labor adjustment costs is less damaging for some aggregate outcomes, but still weakens the fit.
- The baseline estimate of uncertainty shocks assumes high uncertainty is roughly twice low uncertainty, consistent with the volatility spikes in the data.

## 8 Tables and figures

### 8.1 Uncertainty and reduced-form macro evidence

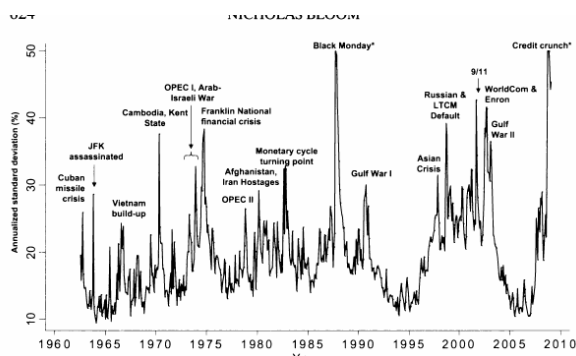


Figure 1. Stock-market volatility spikes after major shocks. The figure motivates the central object of the paper: macro uncertainty shocks.

THE STOCK MARKET VOLATILITY INDEX ROBUSTNESS ON CROSS-SECTIONAL MEASURES OF INCOMPETENCY\*

| Dependent Variable is Stock Market Volatility† | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Stock market volatility, 12-month period       | 0.532 | 0.530 | 0.669 | 0.520 | 0.433 | 0.548 | 0.570 | 0.429 | 0.419 | 0.419 |
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| Stock market volatility, 36-month period       | 0.532 | 0.530 | 0.669 | 0.520 | 0.433 | 0.548 | 0.570 | 0.429 | 0.419 | 0.419 |
| Stock market volatility, 60-month period       | 0.532 | 0.530 | 0.669 | 0.520 | 0.433 | 0.548 | 0.570 | 0.429 |       |       |

Table I. Stock-market volatility is correlated with multiple cross-sectional uncertainty measures, including firm profit dispersion, stock-return dispersion, industry TFP dispersion, and forecast disagreement.

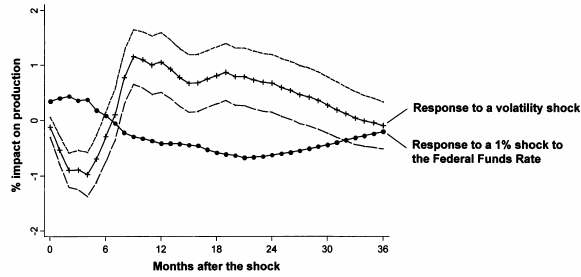


FIGURE 2.—VAR estimation of the impact of a volatility shock on industrial production. *Notes:* Dashed lines are 1 standard-error bands around the response to a volatility shock.

Figure 2. VAR response of industrial production. Industrial production falls rapidly after a volatility shock and then rebounds, unlike the more persistent response to a Federal Funds Rate shock.

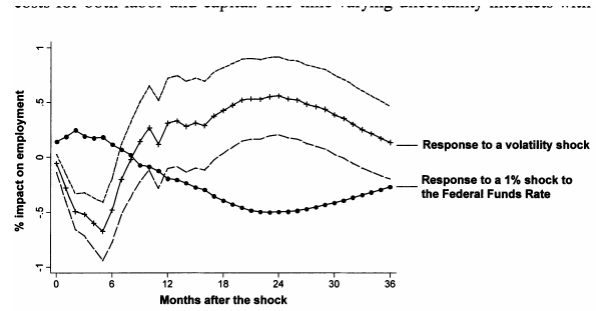


FIGURE 3.—VAR estimation of the impact of a volatility shock on employment. *Notes:* Dashed lines are 1 standard-error bands around the response to a volatility shock.

Figure 3. VAR response of employment. Employment shows the same short-run drop-and-rebound pattern after uncertainty spikes.

## 8.2 Model primitives and policy thresholds

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TABLE II  
PREDEFINED PARAMETERS IN THE MODEL

| Parameter           | Value               | Rationale (Also See the Text)                                                                                                                     |
|---------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha$            | 1/3                 | Capital share in output is one-third, labor share is two-thirds.                                                                                  |
| $\varepsilon$       | 4                   | 33% markup. With constant returns to scale yields $a + b = 0.75$ . I also try a 20% markup to yield $a + b = 0.833$ .                             |
| $w_1$               | 0.8                 | Hourly wages minimized at a 40 hour week.                                                                                                         |
| $w_2$               | 2.4e-9              | Arbitrary scaling parameter. Set so the wage bill equals unity at 40 hours.                                                                       |
| $\sigma_H$          | $2 \times \sigma_L$ | Uncertainty shocks $2 \times$ baseline uncertainty (Figure 1 data). $\sigma_L$ estimated. I also try $1.5 \times$ and $3 \times$ baseline shocks. |
| $\pi_{L,H}^e$       | 1/36                | Uncertainty shocks expected every 3 years (17 shocks in 46 years in Figure 1).                                                                    |
| $\pi_{H,H}^e$       | 0.71                | Average 2-month half-life of an uncertainty shock (Figure 1 data). I also try 1 and 6 month half-lives.                                           |
| $(\mu_H + \mu_L)/2$ | 0.02                | Average real growth rate equals 2% per year. The spread $\mu_H - \mu_L$ is estimated.                                                             |
| $\pi_{L,H}^d$       | $\pi_{H,L}^d$       | Firm-level demand growth transition matrix assumed symmetric. The parameter $\pi_{H,L}^d$ estimated.                                              |

Table II. Predefined model parameters. The table reports production, demand, depreciation, discount-rate, uncertainty, and aggregation assumptions used in the simulations.

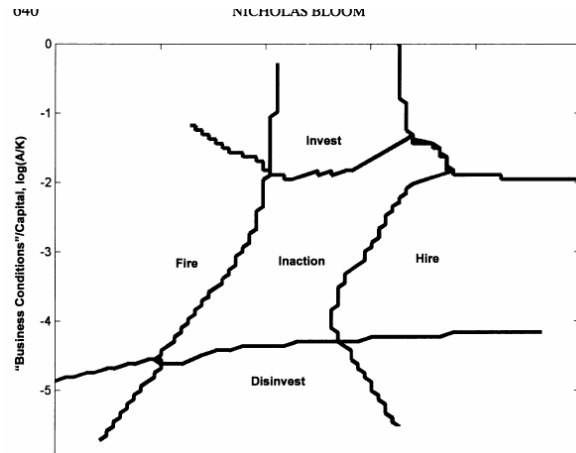


Figure 4. Hiring/firing and investment/disinvestment thresholds. The central inaction region reflects real-options behavior induced by nonconvex adjustment costs.

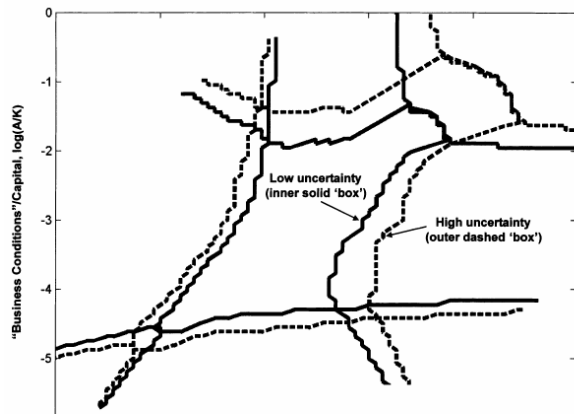


Figure 5. High uncertainty widens the inaction region. Firms require stronger business conditions before hiring or investing when uncertainty is high.

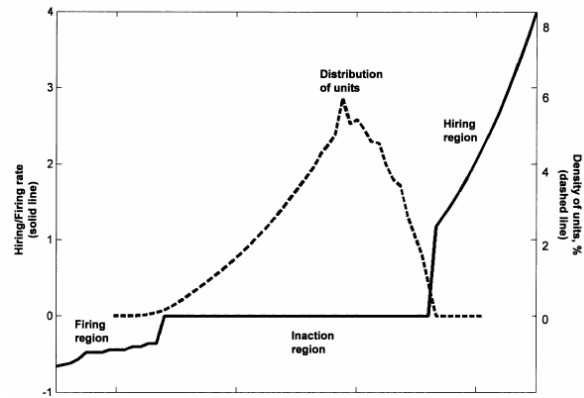


Figure 6. Distribution of units relative to thresholds. Many units are near adjustment thresholds, which helps explain why volatility changes can generate nonlinear aggregate responses.

### 8.3 Baseline uncertainty-shock simulation

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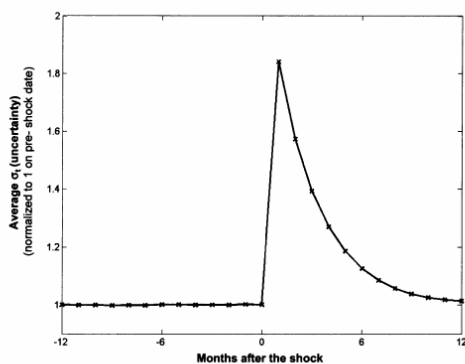


FIGURE 7A.—The simulation has a large second-moment shock.

Figure 7a. The uncertainty shock doubles uncertainty temporarily and then decays quickly.

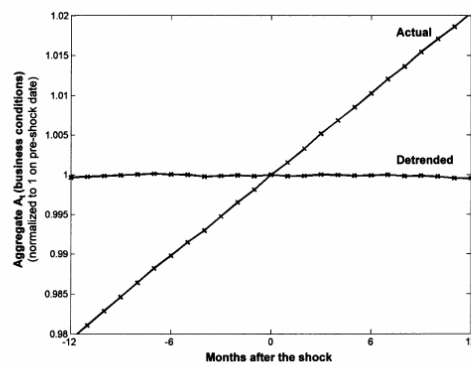


FIGURE 7B.—The simulation has no first moment shock.

Figure 7b. The shock is a pure second-moment shock: average business conditions have no negative first-moment component.

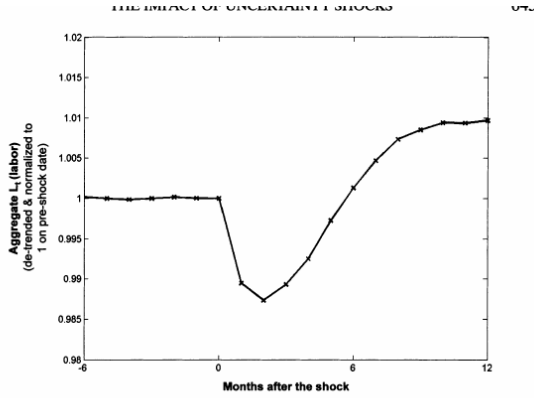


FIGURE 8.—Aggregate (detrended) labor drops, rebounds, and overshoots. The aggregate figures for  $L_t$  are calculated by summing across all units within the simulation.

Figure 8. Aggregate labor drops, rebounds, and overshoots. The early drop comes from delayed hiring, while the recovery reflects pent-up adjustment.

## 8.4 Productivity and output dynamics

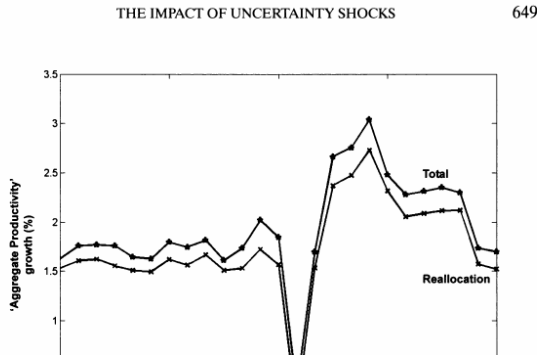


Figure 10a. Productivity growth falls after the shock and then rebounds as reallocation resumes.

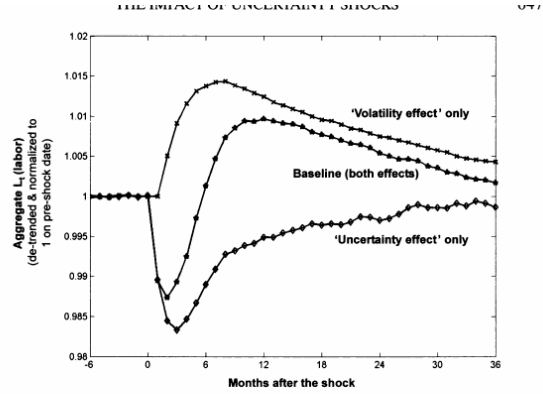


FIGURE 9.—Separating out the uncertainty and volatility effects. The baseline plot is the same as in Figure 8 but extended out for 36 months. For the volatility effect only plot, firms have expectations set to  $\sigma_t = \sigma_L$  in all periods (i.e., uncertainty effects are turned off), while in the uncertainty effect only, they have the actual shocks drawn from a distribution  $\sigma_t = \sigma_L$  in all periods (i.e., the volatility effects are turned off).

Figure 9. Decomposition of uncertainty and volatility effects. The uncertainty effect creates the initial pause; the volatility effect helps generate the later overshoot.

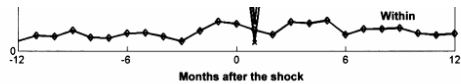


FIGURE 10A.—Aggregate productivity growth falls and rebounds after the shock.

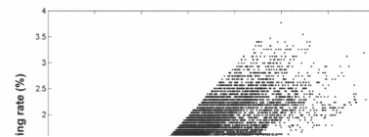


Figure 10b. Before the shock, hiring reallocates resources toward more productive units.

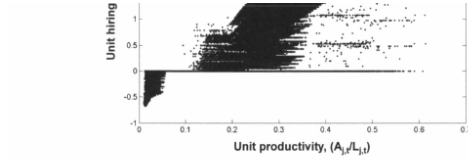


FIGURE 10B.—Unit level productivity and hiring for the period before the uncertainty shock.

Figure 10c. After the shock, hiring is temporarily suppressed, freezing reallocation.

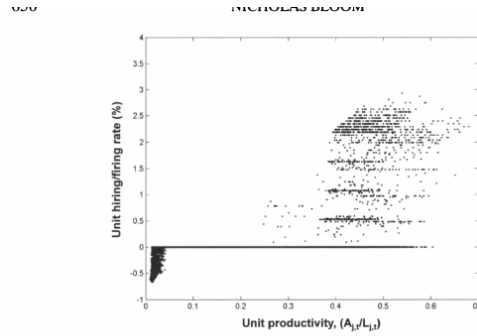


FIGURE 10C.—Unit level productivity and hiring in the period after the uncertainty shock.

In the bottom two panels this reallocation effect is illustrated by two unit-level scatter plots of gross hiring against log productivity in the month before the shock (Figure 10b) and the month after the shock (Figure 10c). It can be seen that after the shock much less reallocation activity takes place with a substan

Figure 11. Solow productivity drops and rebounds, showing that the mechanism affects measured aggregate productivity.

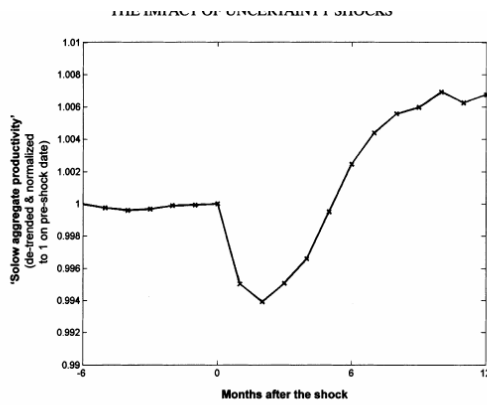


FIGURE 11.—Solow aggregate productivity (detrended) drops, rebounds, and overshoots. Solow productivity is defined as aggregate output divided by the factor share weighted aggregate inputs.

Figure 12. Aggregate output drops, rebounds, and overshoots in the baseline uncertainty-shock simulation.

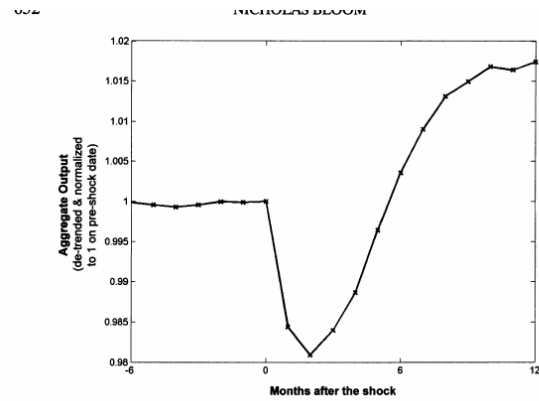


FIGURE 12.—Aggregate (detrended) output drops, rebounds, and overshoots.

Figure 13. VAR response of productivity. The empirical productivity response is consistent with the model's reallocation-freeze mechanism.

This means the results in this model could be compromised by GE effects if factor prices changed sufficiently to counteract factor demand changes.<sup>36</sup> One way to investigate this is to estimate the actual changes in wages, prices, and interest rates that arise after a stock-market volatility shock and feed them into the model in an expectations consistent way. If these empirically plausible changes in factor prices radically changed these results, this would suggest they are not robust to GE, while if they have only a small impact, it is more reassuring on GE robustness.

To do this I use the estimated changes in factor prices from the VAR (see Section 2.2), which are plotted in Figure 13. An uncertainty shock leads to a short-run drop and rebound of interest rates of up to 1.1% points (110 basis point), of prices of up to 0.5%, and of wages of up to 0.3%. I take these numbers and structurally build them into the model so that when  $\sigma_t = \sigma_H$ , interest



Figure 14. Output response with price and wage changes. The short-run effect remains because firms are temporarily insensitive to price incentives when uncertainty is high.

## 8.5 Estimated adjustment costs and robustness

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TABLE III  
ADJUSTMENT COST ESTIMATES (TOP PANEL)\*

| Estimated Parameters                                          | Adjustment Costs Specification |                  |                  |                   |                  |
|---------------------------------------------------------------|--------------------------------|------------------|------------------|-------------------|------------------|
|                                                               | All                            | Capital          | Labor            | Quad              | None             |
| $C_K^e$ : investment resale loss (%)                          | 33.9<br>(6.8)                  | 42.7<br>(14.2)   |                  |                   |                  |
| $C_F^e$ : investment fixed cost (% annual sales)              | 1.5<br>(1.5)                   | 1.1<br>(0.2)     |                  |                   |                  |
| $C_K^Q$ : capital quadratic adjustment cost (parameter)       | 0<br>(0.009)                   | 0.996<br>(0.044) |                  | 4.844<br>(454.15) |                  |
| $C_L^e$ : per capita hiring/firing cost (% annual wages)      | 1.8<br>(0.8)                   |                  | 16.7<br>(0.1)    |                   |                  |
| $C_F^e$ : fixed hiring/firing costs (% annual sales)          | 2.1<br>(0.9)                   |                  | 1.1<br>(0.1)     |                   |                  |
| $C_L^Q$ : labor quadratic adjustment cost (parameter)         | 0<br>(0.037)                   |                  | 1.010<br>(0.017) | 0<br>(0.002)      |                  |
| $\sigma_L$ : baseline level of uncertainty                    | 0.443<br>(0.009)               | 0.413<br>(0.012) | 0.216<br>(0.005) | 0.171<br>(0.005)  | 0.100<br>(0.005) |
| $\mu_H - \mu_L$ : spread of firm business conditions growth   | 0.121<br>(0.002)               | 0.122<br>(0.002) | 0.258<br>(0.001) | 0.082<br>(0.001)  | 0.158<br>(0.001) |
| $\pi_{H,L}^u$ : transition of firm business conditions growth | 0<br>(0.001)                   | 0<br>(0.001)     | 0.016<br>(0.001) | 0<br>(0.001)      | 0.011<br>(0.001) |
| $\gamma$ : curvature of the hours/wages function              | 2.093<br>(0.272)               | 2.221<br>(0.146) | 3.421<br>(0.052) | 2.000<br>(0.009)  | 2.013<br>(14.71) |

(Continues)

Table III, top panel. Baseline adjustment-cost estimates. The preferred specification allows all adjustment costs and fits multiple firm-level moments.

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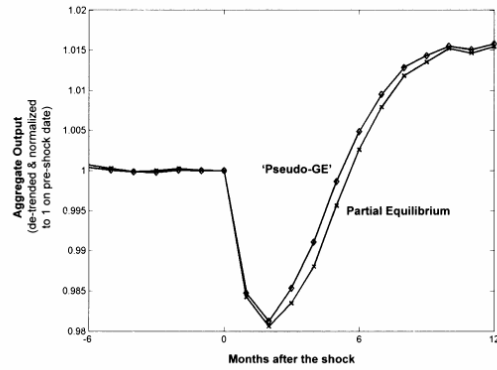


FIGURE 14.—Aggregate (de-trended) output: partial equilibrium and pseudo GE. Pseudo-GE

Figure 15. Combined first- and second-moment shock. Adding a first-moment negative shock creates a deeper and more persistent downturn.

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TABLE III  
(BOTTOM PANEL)

| Moments                                                    | Data  | Data Moments – Simulated Moments |        |        |        |        |
|------------------------------------------------------------|-------|----------------------------------|--------|--------|--------|--------|
| Correlation $(I/K)_{it}$ with $(I/K)_{it-2}$               | 0.328 | 0.060                            | -0.015 | 0.049  | -0.043 | 0.148  |
| Correlation $(I/K)_{it}$ with $(I/K)_{it-4}$               | 0.258 | 0.037                            | 0.004  | 0.088  | 0.031  | 0.162  |
| Correlation $(I/K)_{it}$ with $(\Delta L/L)_{it-2}$        | 0.208 | 0.003                            | -0.025 | 0.004  | -0.056 | 0.078  |
| Correlation $(I/K)_{it}$ with $(\Delta L/L)_{it-4}$        | 0.158 | -0.015                           | -0.009 | 0.036  | 0.008  | 0.091  |
| Correlation $(I/K)_{it}$ with $(\Delta S/S)_{it-2}$        | 0.260 | -0.023                           | -0.062 | -0.044 | -0.102 | 0.024  |
| Correlation $(I/K)_{it}$ with $(\Delta S/S)_{it-4}$        | 0.201 | -0.010                           | -0.024 | 0.018  | -0.036 | 0.087  |
| Standard deviation $(I/K)_{it}$                            | 0.139 | -0.010                           | 0.010  | -0.012 | 0.038  | 0.006  |
| Coefficient of skewness $(I/K)_{it}$                       | 1.789 | 0.004                            | 0.092  | 1.195  | 1.311  | 1.916  |
| Correlation $(\Delta L/L)_{it}$ with $(I/K)_{it-2}$        | 0.188 | -0.007                           | 0.052  | -0.075 | 0.055  | 0.053  |
| Correlation $(\Delta L/L)_{it}$ with $(I/K)_{it-4}$        | 0.133 | -0.021                           | 0.024  | -0.061 | 0.038  | 0.062  |
| Correlation $(\Delta L/L)_{it}$ with $(\Delta L/L)_{it-2}$ | 0.160 | 0.011                            | 0.083  | -0.033 | 0.071  | 0.068  |
| Correlation $(\Delta L/L)_{it}$ with $(\Delta L/L)_{it-4}$ | 0.108 | -0.013                           | 0.054  | -0.026 | 0.045  | 0.060  |
| Correlation $(\Delta L/L)_{it}$ with $(\Delta S/S)_{it-2}$ | 0.193 | -0.019                           | 0.063  | -0.091 | 0.064  | 0.023  |
| Correlation $(\Delta L/L)_{it}$ with $(\Delta S/S)_{it-4}$ | 0.152 | 0.003                            | 0.056  | -0.051 | 0.059  | 0.063  |
| Standard deviation $(\Delta L/L)_{it}$                     | 0.189 | -0.022                           | -0.039 | 0.001  | -0.001 | 0.005  |
| Coefficient of skewness $(\Delta L/L)_{it}$                | 0.445 | -0.136                           | 0.294  | -0.013 | 0.395  | 0.470  |
| Correlation $(\Delta S/S)_{it}$ with $(I/K)_{it-2}$        | 0.203 | -0.016                           | -0.015 | -0.164 | -0.063 | -0.068 |
| Correlation $(\Delta S/S)_{it}$ with $(I/K)_{it-4}$        | 0.142 | -0.008                           | -0.010 | -0.081 | -0.030 | -0.027 |
| Correlation $(\Delta S/S)_{it}$ with $(\Delta L/L)_{it-2}$ | 0.161 | -0.005                           | 0.032  | -0.105 | -0.024 | -0.037 |
| Correlation $(\Delta S/S)_{it}$ with $(\Delta L/L)_{it-4}$ | 0.103 | -0.015                           | 0.011  | -0.054 | -0.005 | -0.020 |
| Correlation $(\Delta S/S)_{it}$ with $(\Delta S/S)_{it-2}$ | 0.207 | -0.033                           | 0.002  | -0.188 | -0.040 | -0.158 |
| Correlation $(\Delta S/S)_{it}$ with $(\Delta S/S)_{it-4}$ | 0.156 | 0.002                            | 0.032  | -0.071 | -0.021 | -0.027 |
| Standard deviation $(\Delta S/S)_{it}$                     | 0.165 | 0.004                            | 0.003  | 0.033  | 0.051  | 0.062  |
| Coefficient of skewness $(\Delta S/S)_{it}$                | 0.342 | -0.407                           | -0.075 | -0.365 | 0.178  | 0.370  |
| Criterion, $\Gamma(\theta)$                                |       | 404                              | 625    | 3618   | 2798   | 6922   |

\*The Data column (bottom panel only) contains the moments from 22,950 observations on investment ( $I$ ), capital ( $K$ ), labor ( $L$ ) and sales ( $S$ ) for 2548 firms. The other columns contain the adjustment costs estimates (top panel) and data moments minus the simulated moments (bottom panel) for all adjustment costs (All), just capital adjustment costs (Capital), just labor adjustment costs (Labor), just quadratic adjustment costs with 1 unit per firm (Quad), and no adjustment costs (None). So, for example, the number 0.328 at the top of the Data column reports that the second-lag of the autocorrelation of investment in the data is 0.328, and the number 0.060 to the right reports that in the All specification the simulated moment is 0.060 less than the data moment (so is 0.268 in total). In the top panel standard-

Table III, bottom panel. Data moments and simulated moments. The model matches investment, labor, and sales dynamics better when both labor and capital frictions are included.

TABLE IV  
A COMPARISON WITH OTHER CAPITAL AND LABOR ADJUSTMENT-COST ESTIMATES

| Source                                 | Capital |           |       | Labor  |           |      |
|----------------------------------------|---------|-----------|-------|--------|-----------|------|
|                                        | PI (%)  | Fixed (%) | Quad  | PI (%) | Fixed (%) | Quad |
| Column All in Table III (this paper)   | 33.9    | 1.5       | 0     | 1.8    | 2.1       | 0    |
| Ramey and Shapiro (2001)               | 40–80   |           |       |        |           |      |
| Caballero and Engel (1999)             |         | 16.5      |       |        |           |      |
| Hayashi (1982)                         |         |           | 20    |        |           |      |
| Cooper and Haltiwanger (2006)          | 2.5     | 20.4      | 0.049 |        |           |      |
| Shapiro (1986)                         |         |           | 3     |        |           | 1.33 |
| Merz and Yashiv (2007)                 |         |           | 14.2  |        |           | 4.2  |
| Hall (2004)                            |         |           | 0     |        |           | 0    |
| Nickell (1986)                         |         |           |       | 8–25   |           |      |
| Cooper, Haltiwanger, and Willis (2004) |         |           |       |        | 1.7       | 0    |

\*PI denotes partial irreversibilities, Fixed denotes fixed costs, and Quad denotes quadratic adjustment costs. Missing values indicate no parameter estimated in the main specification. Zeros indicate the parameter was not significantly different from zero. Nickell's (1986) lower [higher] value is for unskilled [skilled] workers. Shapiro's (1986) value is a weighted average of  $(2/3) \times 0$  for production workers and  $(1/3) \times 4$  for nonproduction workers. Merz and Yashiv's (2007) values are the approximated quadratic adjustment costs at the sample mean. Comparability subject to variation in data sample, estimation technique, and maintained assumptions.

Table IV. Comparison with previous adjustment-cost estimates. Bloom's estimates are broadly consistent with prior work but are jointly estimated for labor and capital.

are lower. The column  $a + b = 0.83$  reports the results for a specification with a 20% markup, in which the estimated adjustment costs look very similar to the baseline results.

In columns  $N = 25$  and  $N = 1$ , the results are reported for simulations assuming the firm operates 25 units and 1 unit, respectively.<sup>31</sup> These assumptions also lead to higher estimates for the quadratic adjustment costs and lower estimates for the nonconvex adjustment costs to compensate for the reduction in smoothing by aggregation.

In columns Pre -5% and Pre +5%, the results are reported for simulations assuming there is a -5% and +5% shock, respectively, to business conditions in the 6 months before the simulation begins. This simulates running the simulation during a recession and boom, respectively, to investigate how the initial conditions influence the results. As shown in Table V the results are numerically identical for Pre -5% and similar for Pre +5% to those for All, suggesting the results are robust to different initial conditions. The reason is that the long time period of the simulation (20 years) and the limited persistence of macro shocks means that the impact of initial conditions dies quickly.

In column Cap ME, the parameters are estimated including a log-normal 10% measurement error for capital, with broadly similar results. Finally, the last two columns investigate the impact of time aggregation. First, the column Yearly reports the results for running the simulation at a yearly frequency with-

Figure 17. Parameter robustness. The qualitative drop-rebound pattern survives alternative production and adjustment assumptions.

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Adjustment Costs Specification

|         |         | Pre -5% |         |         | Pre +5% |         |  | Cap ME |  |  | Yearly |  |  | Weekly |  |  |
|---------|---------|---------|---------|---------|---------|---------|--|--------|--|--|--------|--|--|--------|--|--|
| N = 25  | N = 1   |         |         |         |         |         |  |        |  |  |        |  |  |        |  |  |
| 30.3    | 47.0    | 33.9    | 36.1    | 35.3    | 45.3    | 50.0    |  |        |  |  |        |  |  |        |  |  |
| (8.7)   | (9.1)   | (6.8)   | (4.9)   | (5.1)   | (5.2)   | (30.8)  |  |        |  |  |        |  |  |        |  |  |
| 0.9     | 1.3     | 1.5     | 1.6     | 0.9     | 2.1     | 1.6     |  |        |  |  |        |  |  |        |  |  |
| (0.4)   | (0.2)   | (1.0)   | (0.5)   | (0.4)   | (0.3)   | (0.4)   |  |        |  |  |        |  |  |        |  |  |
| 0.616   | 2.056   | 0       | 0.058   | 0.525   | 0.025   | 1.488   |  |        |  |  |        |  |  |        |  |  |
| (0.154) | (0.284) | (0.009) | (0.037) | (0.078) | (0.015) | (0.729) |  |        |  |  |        |  |  |        |  |  |
| 0       | 0       | 1.8     | 0       | 0       | 2.0     | 0       |  |        |  |  |        |  |  |        |  |  |
| (0.1)   | (0.1)   | (0.8)   | (0.04)  | (0.7)   | 0.9     | (6.1)   |  |        |  |  |        |  |  |        |  |  |
| 1.3     | 0       | 2.1     | 1.6     | 0.9     | 2.0     | 1.3     |  |        |  |  |        |  |  |        |  |  |
| (0.8)   | (0.0)   | (0.9)   | (0.7)   | (0.4)   | (0.5)   | (0.3)   |  |        |  |  |        |  |  |        |  |  |
| 0.199   | 0.070   | 0       | 0       | 0       | 1.039   | 0.808   |  |        |  |  |        |  |  |        |  |  |
| (0.062) | (0.031) | (0.037) | (0.018) | (0.017) | (0.163) | (0.612) |  |        |  |  |        |  |  |        |  |  |
| 0.393   | 0.248   | 0.443   | 0.469   | 0.515   | 0.339   | 0.600   |  |        |  |  |        |  |  |        |  |  |
| (0.013) | (0.008) | (0.009) | (0.011) | (0.017) | (0.011) | (0.035) |  |        |  |  |        |  |  |        |  |  |
| 0.163   | 0.126   | 0.121   | 0.132   | 0.152   | 0.228   | 0       |  |        |  |  |        |  |  |        |  |  |
| (0.002) | (0.002) | (0.002) | (0.002) | (0.003) | (0.005) | (0.018) |  |        |  |  |        |  |  |        |  |  |
| 0       | 0       | 0       | 0       | 0       | 0.016   | n/a     |  |        |  |  |        |  |  |        |  |  |
| (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) |  |        |  |  |        |  |  |        |  |  |
| 2.148   | 2.108   | 2.093   | 2.056   | 2.000   | 2.000   | 2.129   |  |        |  |  |        |  |  |        |  |  |
| (0.260) | (0.147) | (0.272) | (0.264) | (0.211) | (0.166) | (0.254) |  |        |  |  |        |  |  |        |  |  |
| 556     | 593     | 403     | 380     | 394     | 656     | 52      |  |        |  |  |        |  |  |        |  |  |

a model (All), except 20% annualized labor attrition ( $\delta_L = 20\%$ ), 20% markup in business shock  $\epsilon$  enters prior to the start of the simulation period (Pre -5%), a 10% measurement error for capital (Cap ME), and quadratic adjustment costs (Quad). The results are reported for simulations assuming the firm operates 25 units and 1 unit, respectively. These assumptions also lead to higher estimates for the quadratic adjustment costs and lower estimates for the nonconvex adjustment costs to compensate for the reduction in smoothing by aggregation. It reports parameters estimated using moment from simulated quarterly, and then aggregating up to yearly data. This is done to test the bias from generated from an underlying weekly process. Standard errors are given in italics.

Figure 16. Robustness to different adjustment-cost structures. The short-run uncertainty mechanism depends on nonconvex adjustment costs.

### 6.1. Adjustment Costs

To evaluate the effects of different types of adjustment, I ran three simulations: the first with fixed costs only, the second with partial irreversibilities only, and the third with quadratic adjustment costs only.<sup>32</sup> The output from these three simulations is shown in Figure 16. As can be seen, the two specifications with nonconvex adjustment costs generate a distinct drop and rebound in economic activity. The rebound with fixed costs is faster than with partial irreversibilities because of the bunching in hiring and investment, but otherwise they are remarkably similar in size, duration, and profile. The quadratic adjustment-cost specification appears to generate no response to an uncertainty shock. The reason is that there is no kink in adjustment costs around zero, so there is no option value associated with doing nothing.

In summary, this suggests the predictions are very sensitive to the inclusion of some degree of nonconvex adjustment costs, but are much less sensitive to the type (or indeed level) of these nonconvex adjustment costs. This highlights

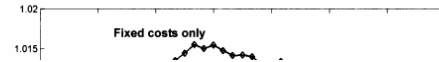


Figure 18. Uncertainty-shock half-lives. More persistent uncertainty generates a more prolonged downturn.

the importance of the prior step of estimating the size and nature of the underlying labor and capital adjustment costs.

### 6.2. Predefined Parameters

To investigate the robustness of the simulation results to the assumptions over the predefined parameters, I reran the simulations using the different parameters from Table V. The results, shown in Figure 17, highlight that the qualitative result of a drop and rebound in activity is robust to the different assumptions over the predetermined parameters. This is because of the presence of some nonconvex component in all the sets of estimated adjustment costs in Table V.

The size of this drop and rebound did vary across specifications, however. Running the simulation with the  $N = 1$  parameter estimates from Table V leads to a drop of only 1%, about half the baseline drop of about 1.8%. This smaller drop was due to the very high levels of estimated quadratic adjustment costs that were required to smooth the investment and employment series in the absence of cross-sectional aggregation. Of course, the assumption of no cross-sectional aggregation ( $N = 1$ ) is inconsistent with the pervasive aggregation in the typical large firm. This simulation is presented simply to highlight

Figure 19. Shock size robustness. Larger uncertainty shocks generate larger drops and rebounds.

In the  $\delta_L = 0.2$  specification the drop was around 2.25%, about 30% above the baseline drop, due to the greater labor attrition after the shock. Hence, this more realistic assumption on 20% annual labor attrition (rather than 10% in the baseline) generates a larger drop and rebound in activity. The results for assuming partial cross-sectional aggregation ( $N = 25$ ), a 20% markup ( $a + b = 0.83$ ), a preestimation boom (Pre +5%), and capital measurement errors (Cap ME) are all pretty similar to the baseline simulation (which has full cross-sectional aggregation and a 33% markup).

### 6.3. Durations and Sizes of Uncertainty Shocks

Finally, I also evaluate the effects of robustness of the simulation predictions to different durations and sizes of uncertainty shocks. In Figure 18, I plot the output response to a shorter-lived shock (a 1-month half-life) and a longer-lived shock (a 6-month half-life). Also plotted is the baseline (a 2-month half-life). It is clear that longer-lived shocks generate larger and more persistent falls in output. The reason is that the pause in hiring and investment lasts for longer if the rise in uncertainty is more persistent. Of course, because the rise in uncertainty is more persistent, the cumulative increase in volatility is also larger so that the medium-term “volatility overshoot” is also greater. Hence, *more persistent uncertainty shocks generate a larger and more persistent drop*

Figure 20. Macro uncertainty only. Even when only aggregate uncertainty changes, the model still produces the characteristic uncertainty-shock response.

## 9 How to read the key tables and figures

- **Figure 1 and Table I:** These establish the measurement argument. Stock-market volatility is not merely a financial-market object; it moves with broader uncertainty measures.
- **Figures 2 and 3:** These show the empirical macro pattern. The economy responds to uncertainty shocks with a sharp but temporary contraction.
- **Figures 4 to 6:** These provide the real-options intuition. Adjustment thresholds and inaction regions are the micro foundation for aggregate nonlinearity.
- **Figures 7 to 15:** These are the central simulation results. They show how a pure second-moment shock affects labor, productivity, and output.
- **Tables III and IV:** These discipline the adjustment-cost assumptions and show that the estimated frictions are plausible relative to existing literature.
- **Figures 16 to 20:** These show that the mechanism is robust but quantitatively sensitive to adjustment costs, uncertainty persistence, and shock size.

## 10 Economic intuition

- In normal times, firms near their hiring or investment thresholds adjust when conditions improve enough.
- After an uncertainty shock, the value of waiting increases because adjustment may become costly to reverse.
- This temporary pause reduces current hiring, investment, and reallocation.
- Once uncertainty falls, delayed activity resumes. The rebound is strong because firms have accumulated pent-up demand.
- If volatility remains elevated, convexity near the thresholds can create an overshoot in labor, output, and productivity.

## 11 Limitations

- The baseline model abstracts from full general equilibrium feedbacks, although the paper conducts robustness exercises with wage, price, and interest-rate responses.
- The uncertainty proxy is stock-market volatility, which may include risk premia, financial conditions, and other market-specific forces.
- The model assumes a particular adjustment-cost structure and stochastic process; alternative firm dynamics could change quantitative magnitudes.
- The model uses Compustat firms, which are large publicly listed firms and may not represent small private firms.
- Entry and exit are not central in the baseline model, even though uncertainty may also affect business formation and shutdown decisions.

## 12 Conclusion

- Bloom (2009) shows that uncertainty shocks can independently generate short, sharp recessions and recoveries.
- The key mechanism is the interaction between time-varying uncertainty and nonconvex adjustment costs: uncertainty raises the option value of waiting.
- The paper's most important contribution is to connect micro-level real-options behavior to aggregate fluctuations in employment, output, and productivity.
- For policy, the paper warns that after uncertainty spikes, firms may be temporarily less responsive to conventional stimulus because they are waiting for uncertainty to resolve.